LearnHub

Lecture notes: by date and by subject (minor differences may occur between the two versions)

Class exercises and their solutions

Reading assignments

- Reading assignment 1: Chapter 1 (Assessments) due this Friday. → will be extended due to the problems in getting the connect codes.
- You will have one week to work on each reading assignment.

Labs:

- Lab 3 material is accessible since yesterday

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Today

Class exercise and propagation of error (Sections 3.1 and 3.2)

Point estimators (Section 4.9)

Lab 2 review: this Thursday

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Class exercise

Consider the random variables X and Y with the following joint probability mass function:

$P_{X,Y}(1,1) = P_{X,Y}(0,0) = \frac{3}{8}$; $P_{X,Y}(0,1) = P_{X,Y}(1,0) = \frac{1}{8}$

a) (E[X], E[Y])=?

b) (Var(X), Var(Y))=?

c) Cov(X,Y)=?

d) Var(2X+3Y)=?

$Var(X) = E[(X - \mu_X)^2] = E[X^2] - (\mu_X)^2$
$Cov(X,Y) = E[(X - \mu_X)(Y - \mu_Y)] = E[XY] - \mu_X \mu_Y$
$Var(aX + bY) = a^2 Var(X) + b^2 Var(Y) + 2abCov(X,Y)$

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Point estimators (Section 4.9)

Let $X_1,\ldots,X_n$ be a (simple) random sample.

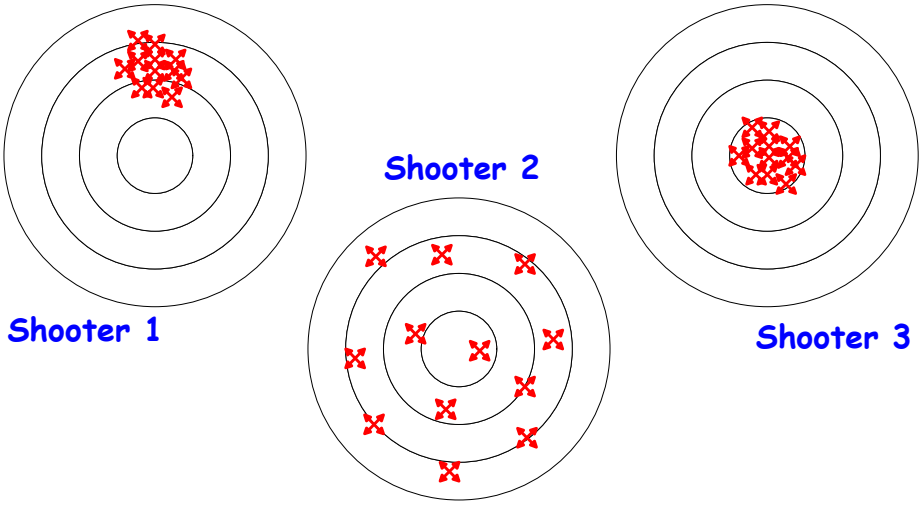
A statistic is any function of the observations in a random sample.

A statistic is a random variable, so it has its own probability distribution (generally called sampling distribution).

A statistic that is used to estimate the value of a parameter is called point estimator.

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Which shooter would you hire if you were the chief officer of the police station?



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Point Estimation

Target → Population parameter θ
(such as mean, variance, median etc.)

Shot → Point estimate of a parameter θ
(a sensible value for θ, a single number)

Shooter → Point estimator of a parameter θ
(a rule, mostly a formula, to calculate an estimate based on the sample)

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Point Estimation

Let X be a random variable with probability distribution f(x; θ) where θ is an unknown parameter.

Let $X_1, X_2, \ldots, X_n$ be a random sample from X.

$\Rightarrow$ the statistic = $h(X_1, X_2, \ldots, X_n)$ corresponding to θ is the estimator of θ.

Note: The estimator is itself a random variable since it is based on sample.

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Simple questions

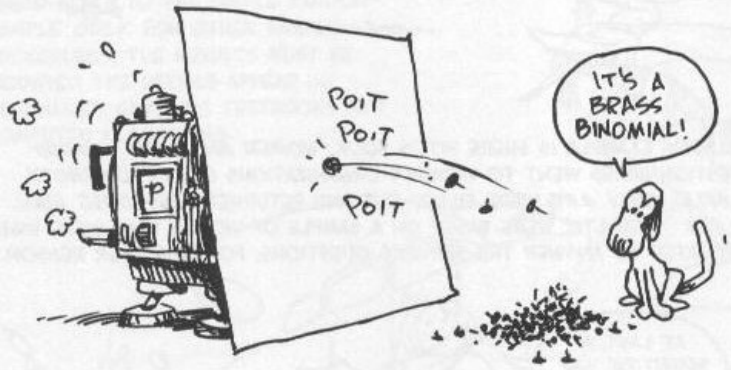
$X_1, X_2, \ldots, X_n$ be a random sample from the random variable X.

- How to estimate the mean of X from this sample?
- How to estimate the standard deviation of X from this sample?
- How to estimate the median of X from this sample?

Other simple questions…

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THE ASTUTE READER WILL RECOGNIZE THIS AS A ***BERNOULLI SYSTEM:*** EACH NEW TACK IS THE OUTCOME OF A BERNOULLI TRIAL WITH SOME PROBABILITY p OF SUCCESS (I.E., BEING DEFECT-FREE) AND PROBABILITY $1-p$ OF FAILURE (I.E., BEING DEFECTIVE).



WE THINK OF THIS SITUATION AS IF THERE WERE A ***HIDDEN BUT REAL "BERNOULLI MACHINE"*** WHOSE PROBABILITY p GOVERNS THE OUTCOMES WE OBSERVE IN THE SO-CALLED "REAL WORLD."

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TO BE PRECISE, IF X IS THE NUMBER OF SUCCESSES IN THE SAMPLE, THEN X IS NOTHING BUT OUR OLD FRIEND THE BINOMIAL RANDOM VARIABLE (n TRIALS, PROBABILITY p)... AND WE DEFINE THE ***OBSERVED PROPORTION*** TO BE THE RANDOM VARIABLE

$$\hat{P} = \frac{X}{n}$$



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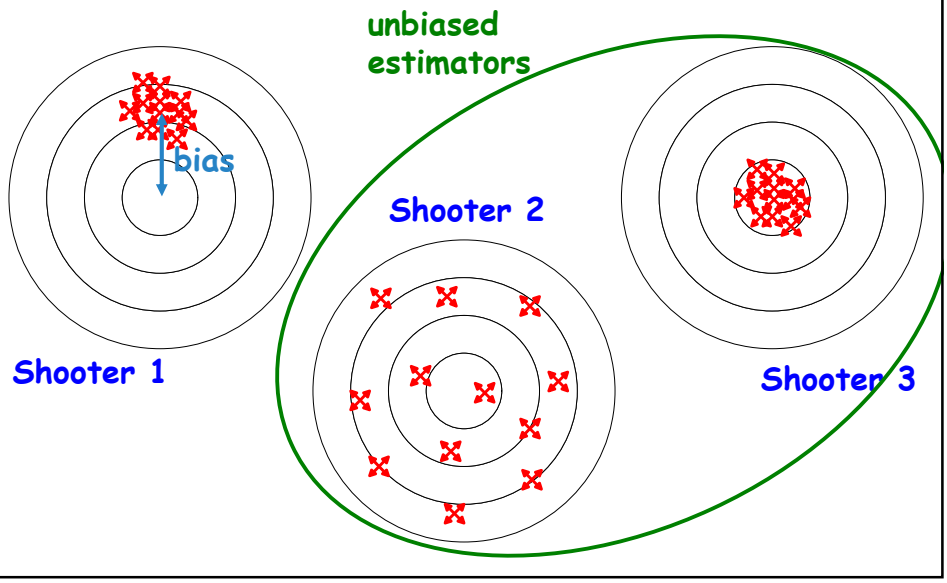
Unbiased estimators

Definition: A point estimator $\hat{\theta}$ is said to be an unbiased estimator of θ if $\mathrm{E}[\hat{\theta}] = \theta$ for all possible values of θ.

If $\hat{\theta}$ is not unbiased, $\mathrm{E}[\hat{\theta}] - \theta$ is called the bias of $\hat{\theta}$.

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Which shooter would you hire if you were the chief officer of the police station?



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Mean Square Error of an Estimator

Definition

The **mean square error** of an estimator $\hat{\Theta}$ of the parameter θ is defined as

$$\mathrm{MSE}(\hat{\Theta}) = E(\hat{\Theta} - \theta)^2$$

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Mean Square Error of an Estimator

The **mean square error** of an estimator $\hat{\Theta}$ of the parameter θ is defined as

$$\text{MSE}(\hat{\Theta}) = E(\hat{\Theta} - \theta)^2 \qquad (7\text{-}3)$$

$$MSE(\hat{\Theta}) = E[(\hat{\Theta} - \theta)^2] = E[(\hat{\Theta} - E[\hat{\Theta}] + E[\hat{\Theta}] - \theta)^2]$$

$$= E[(\hat{\Theta} - E[\hat{\Theta}])^2 + (E[\hat{\Theta}] - \theta)^2 + 2(\hat{\Theta} - E[\hat{\Theta}])(E[\hat{\Theta}] - \theta)]$$

$$= E[(\hat{\Theta} - E[\hat{\Theta}])^2] + E[(E[\hat{\Theta}] - \theta)^2] + 2E[(\hat{\Theta} - E[\hat{\Theta}])(E[\hat{\Theta}] - \theta)]$$

$$= E\left[(\hat{\Theta} - E[\hat{\Theta}])^2\right] + (E[\hat{\Theta}] - \theta)^2 + 2(E[\hat{\Theta}] - \theta)E[(\hat{\Theta} - E[\hat{\Theta}])]$$

Variance (Bias)^2 0

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Mean Square Error of an Estimator

$$\text{MSE}_{\hat{\theta}} = (\mu_{\hat{\theta}} - \theta)^2 + (\sigma_{\hat{\theta}})^2$$

standard error of $\hat{\theta}$

Definition: $\sigma_{\hat{\theta}} = \sqrt{V(\hat{\theta})}$ is called the standard error of an estimator $\hat{\theta}$.

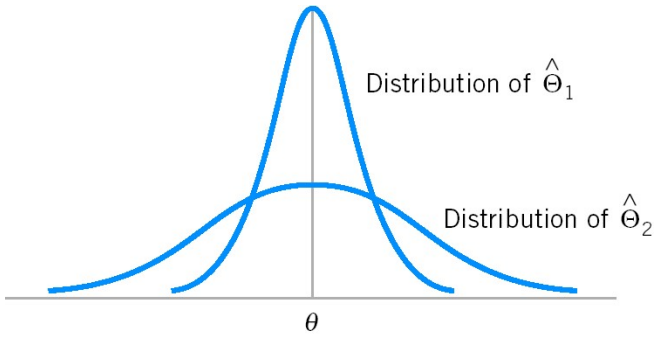
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General Concepts of Point Estimation

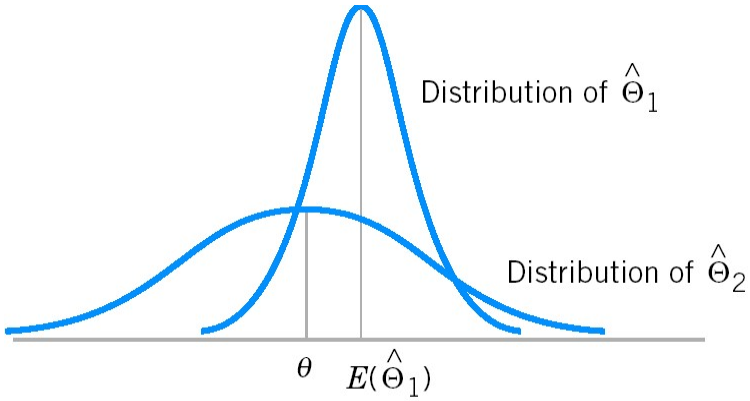
Variance of a Point Estimator

The sampling distributions of two unbiased estimators $\hat{\Theta}_1$ and $\hat{\Theta}_2$.



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Mean Square Error of an Estimator



A biased estimator $\hat{\Theta}_1$ that has smaller variance than the unbiased estimator $\hat{\Theta}_2$.

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Which estimator is unbiased?

Definition: A point estimator $\hat{\theta}$ is said to be an unbiased estimator of θ if $\mathrm{E}[\hat{\theta}] = \theta$ for all possible values of θ.

If $\hat{\theta}$ is not unbiased $\mathrm{E}[\hat{\theta}] - \theta$ is called the bias of $\hat{\theta}$.

Is $\hat{p}$ unbiased? Can we find its MSE?

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What else is unbiased?

Proposition: Let X be a random variable with mean μ & variance σ^2. Let $X_1, \ldots, X_n$ be a random sample of size *n* from X.

a. $\bar{X}$ is an unbiased estimator of μ.

MSE of $\bar{X}$: ?

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